In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df=pd.read_csv("Iris.csv")
df.tail()
#df['Species'].unique()
```

Out[2]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

In [3]:
```
df.isnull().sum()
```

Out[3]:
```
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

In [4]:
```
df.dtypes
```

Out[4]:
```
Id                 int64
SepalLengthCm    float64
SepalWidthCm     float64
PetalLengthCm    float64
PetalWidthCm     float64
Species           object
dtype: object
```

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

In [6]:

```
df.describe()
```

Out[6]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

In [7]:

```
'' 'Iris-setosa' 'Iris-virginica' 'Iris-versicolor'
```

Out[7]:

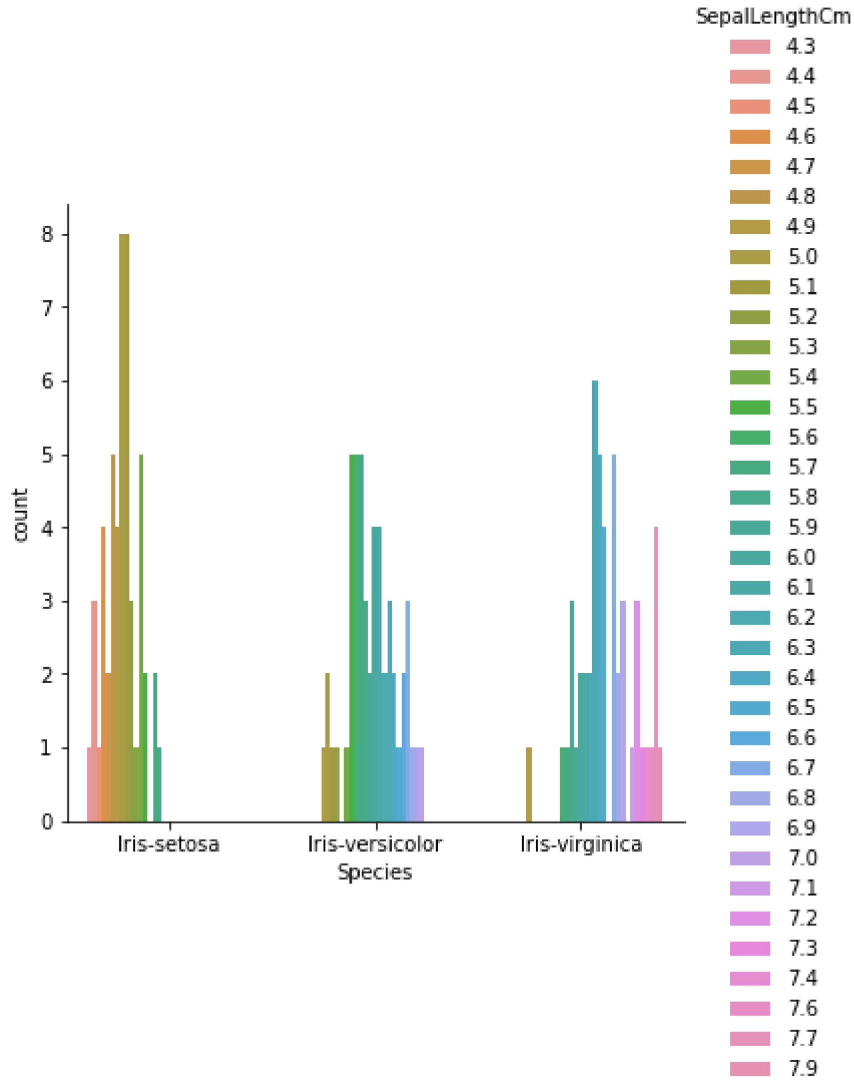
```
'Iris-setosaIris-virginicaIris-versicolor'
```

In [8]:

```
sns.catplot(x ="Species", hue ="SepalLengthCm",
kind ="count", data = df)
```

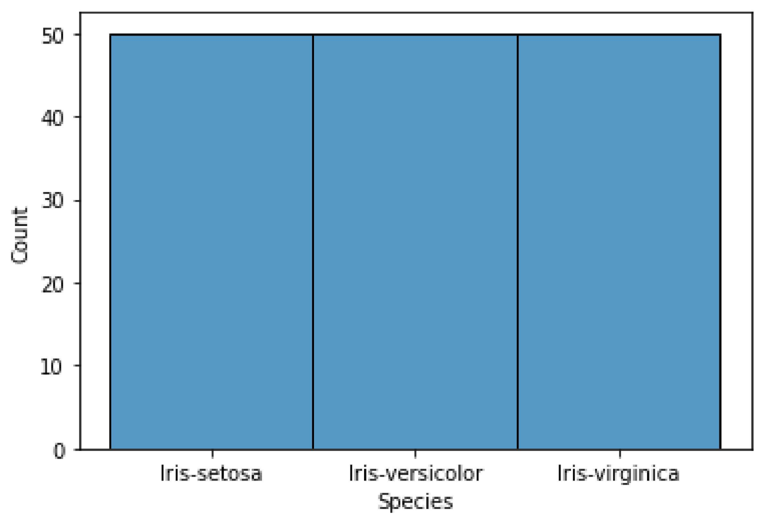
Out[8]:

```
<seaborn.axisgrid.FacetGrid at 0x7fd501a8da30>
```



In [9]:
```
sns.histplot(data=df, x="Species")
```

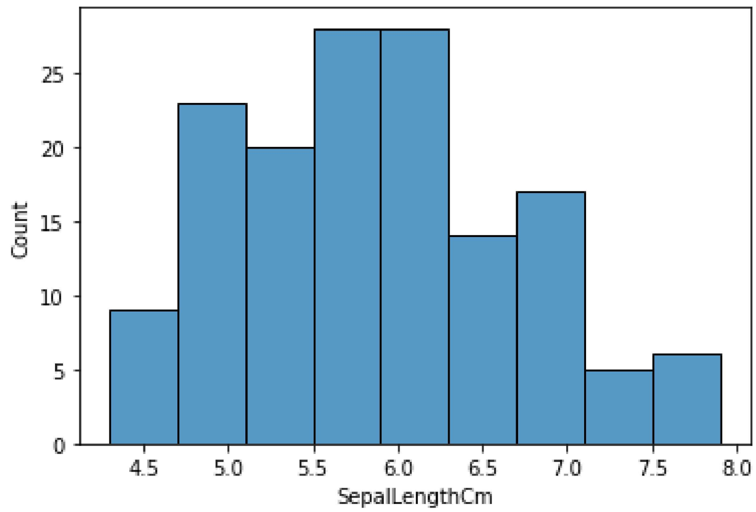
Out[9]: <AxesSubplot:xlabel='Species', ylabel='Count'>



In [10]:
```
sns.histplot(data=df, x="SepalLengthCm")
#Average sepallength lies between 5 -6
```

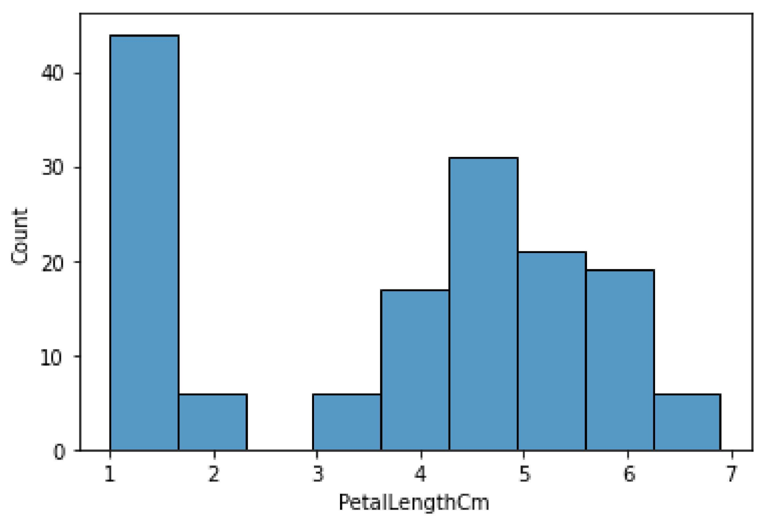
```
df['SepalLengthCm'].mean()
```

Out[10]: 5.843333333333334



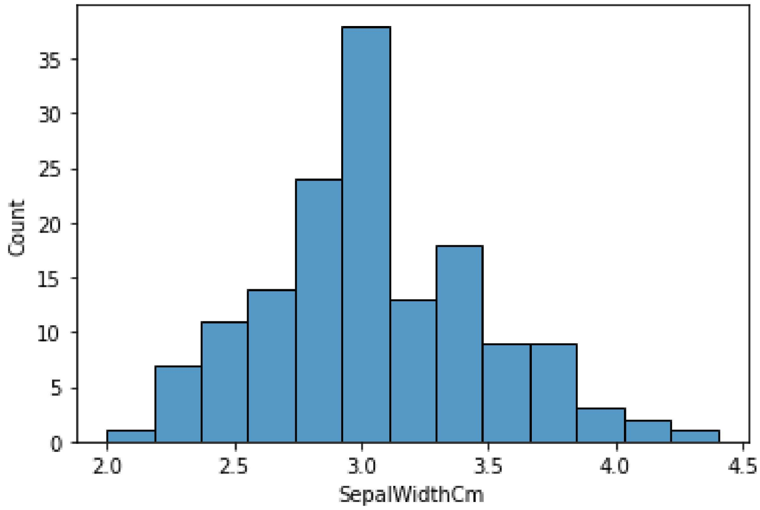
In [11]:
```
sns.histplot(data=df, x="PetalLengthCm")
#Average sepalwidth lies between 4 -6
df['PetalLengthCm'].mean()
```

Out[11]: 3.758666666666666



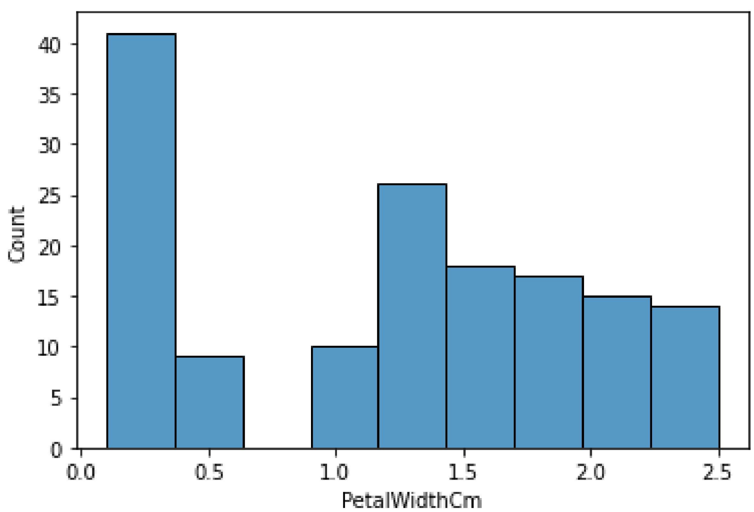
In [12]:
```
sns.histplot(data=df, x="SepalWidthCm")
#Average sepalwidth lies between 2.5 -3.5
df['SepalWidthCm'].mean()
```

Out[12]: 3.0540000000000003



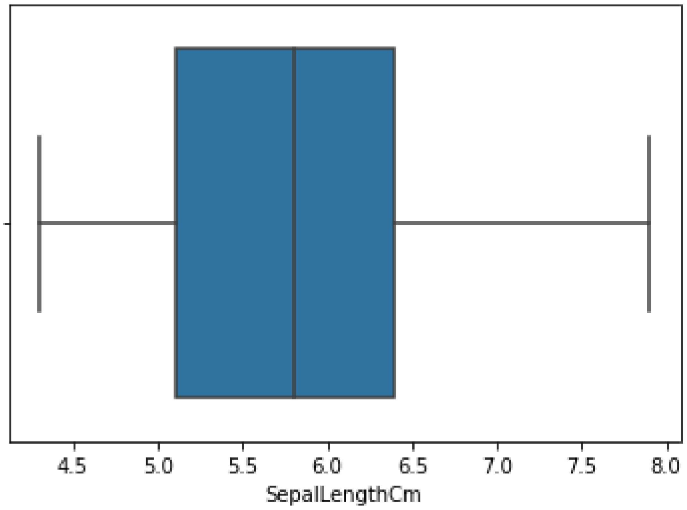
In [13]:
```
sns.histplot(data=df, x="PetalWidthCm")
#Average petalwidth lies between 1 -2
df['PetalWidthCm'].mean()
```

Out[13]: 1.1986666666666668



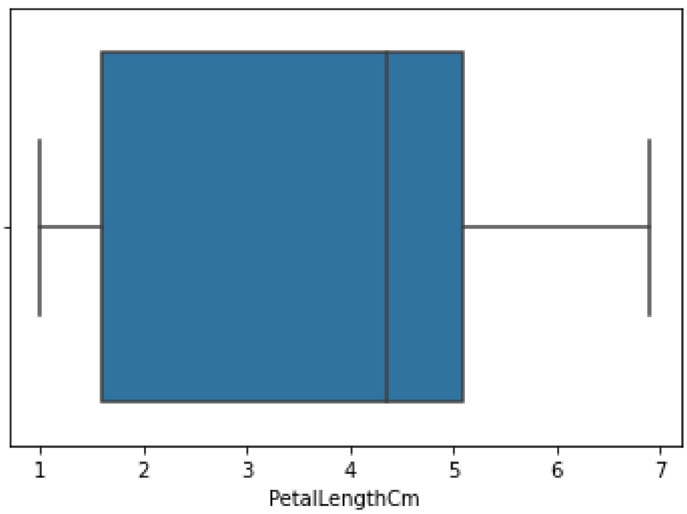
In [14]:
```
sns.boxplot(x=df['SepalLengthCm'])
#Median lies between 5.5 to 6
```

Out[14]: <AxesSubplot:xlabel='SepalLengthCm'>



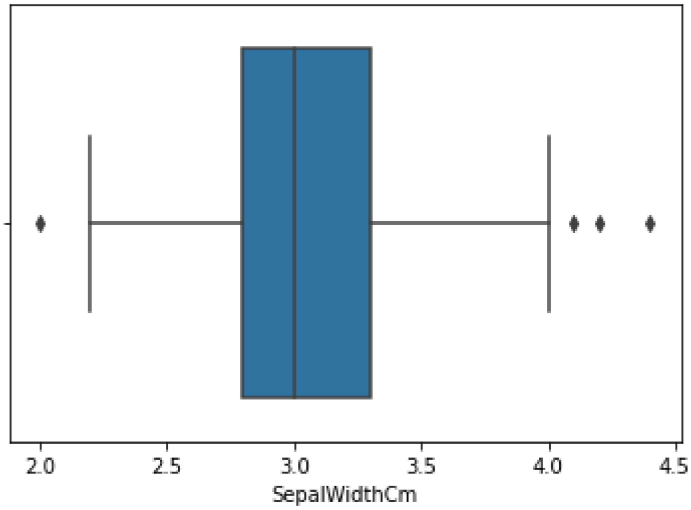
In [15]:
```
sns.boxplot(x=df['PetalLengthCm'])
#Median lies between 4 to 5
```

Out[15]: <AxesSubplot:xlabel='PetalLengthCm'>



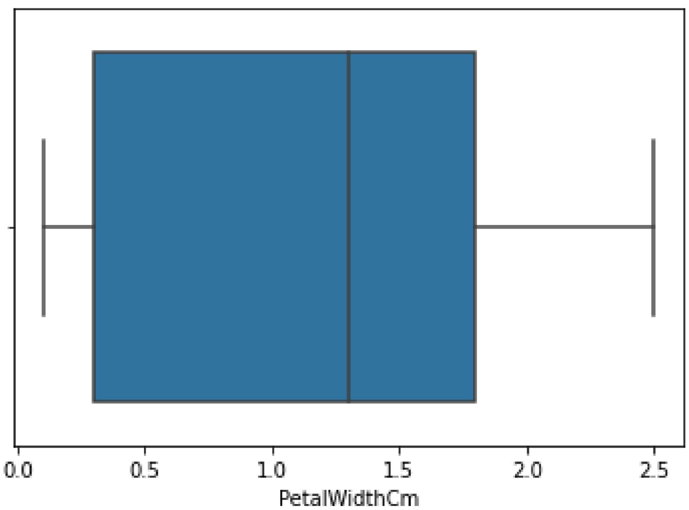
In [16]:
```
sns.boxplot(x=df['SepalWidthCm'])
```

Out[16]: <AxesSubplot:xlabel='SepalWidthCm'>



In [17]:
```
sns.boxplot(x=df['PetalWidthCm'])
#Median lies between 1 to 1.5
```

Out[17]: <AxesSubplot:xlabel='PetalWidthCm'>



In [18]:
```
Q1=df['SepalWidthCm'].quantile(0.25)
Q3=df['SepalWidthCm'].quantile(0.75)
Q1,Q3
```

Out[18]: (2.8, 3.3)

In [19]:
```
IQR=Q3-Q1
```

In [20]:
```
lower_limit=Q1-1.5*IQR
upper_limit=Q3+1.5*IQR
```

In [21]:
```
df[(df['SepalWidthCm']<lower_limit)|(df['SepalWidthCm']>upper_limit)]
```

Out[21]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
15	16	5.7	4.4	1.5	0.4	Iris-setosa
32	33	5.2	4.1	1.5	0.1	Iris-setosa
33	34	5.5	4.2	1.4	0.2	Iris-setosa
60	61	5.0	2.0	3.5	1.0	Iris-versicolor

In [22]:
```
df_without_outliers=df[(df['SepalWidthCm']>lower_limit)&(df['SepalWidthCm']<upper_l
df_without_outliers
```

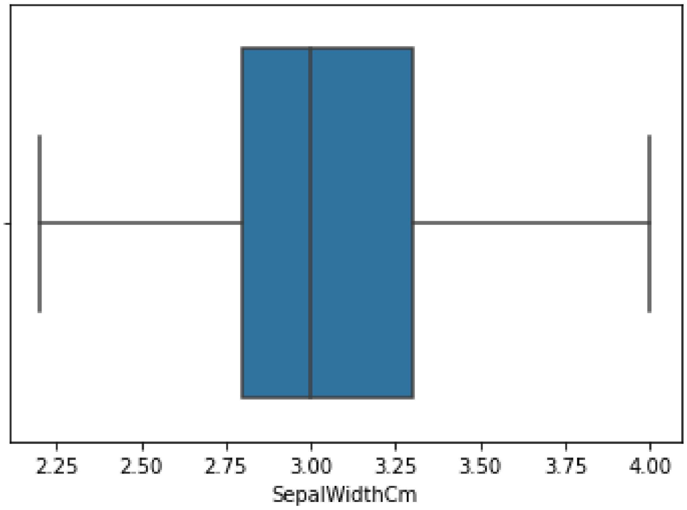
Out[22]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

146 rows × 6 columns

In [23]:
```
sns.boxplot(x=df_without_outliers['SepalWidthCm'])
```

Out[23]: <AxesSubplot:xlabel='SepalWidthCm'>



In [24]:
```
df.corr()
```

Out[24]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Id	1.000000	0.716676	-0.397729	0.882747	0.899759
SepalLengthCm	0.716676	1.000000	-0.109369	0.871754	0.817954
SepalWidthCm	-0.397729	-0.109369	1.000000	-0.420516	-0.356544
PetalLengthCm	0.882747	0.871754	-0.420516	1.000000	0.962757
PetalWidthCm	0.899759	0.817954	-0.356544	0.962757	1.000000

In [25]:
```
features = df.iloc[:, 1:5]
```

In [26]:
```
target = df.iloc[:, -1]
```

In [27]:
```
target.head()
```

Out[27]:
```
0    Iris-setosa
1    Iris-setosa
2    Iris-setosa
3    Iris-setosa
4    Iris-setosa
Name: Species, dtype: object
```

In [28]:
```
print('The Initial DataFrame Contained %d Rows And %d Columns'%(df.shape))

print('The Features Matrix Contains %d Rows And %d Columns'%(features.shape))

print('The Target Vector Contains %d Rows And %d Columns'%(np.array(target).reshape
```

```
The Initial DataFrame Contained 150 Rows And 6 Columns
The Features Matrix Contains 150 Rows And 4 Columns
The Target Vector Contains 150 Rows And 1 Columns
```